

COMPUTER PRACTICE 6.

MULTIPLE LINEAR REGRESSION

OBJECTIVES

- Acquire practical skills in multiple regression analysis
- Learn how to estimate unknown multiple linear regression model parameters
- Learn how to perform goodness-of-fit test for multiple linear regression model
- Learn how to apply stepwise procedures for multiple linear regression model construction

GUIDELINES

- Load the data according to your option (ask your tutor).
- Construct the multiple regression and analyse the result:
 - Model's quality parameters
 - Model coefficients significance analysis using t-Student test
 - Residual analysis (Gauss-Markov condition).
- Check the Gauss-Markov condition for residuals. In case of outliers, consistently (one by one) exclude outliers from the model.
- After removing of outliers, exclude insignificant variables from the model (one by one).
- Build non-linear models using two the most significant predictors:
$$y = b_0 + b_1x_1^2 + b_2x_2^2 + e;$$
$$y = b_0 + b_1\ln(x_1) + b_2\ln(x_2) + e.$$
- Use stepwise methods to find the best regression model. Use both, Forward stepwise and Backward stepwise.
- Copy the final result to the report as shown at Figure 1.

```

Call:
lm(formula = ARREAR_Y ~ CITY + EQUITY + GOVLIAB + PERF + PROTECT,
    data = dat)

Residuals:
    Min       1Q   Median       3Q      Max
-10070.8  -5092.0   -563.3   3821.6  20491.1

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.734e+03  4.442e+03   0.615  0.54340
CITY        -5.760e+03  1.677e+03  -3.434  0.00193 **
EQUITY       -6.959e-01  2.317e-01  -3.004  0.00570 **
GOVLIAB     -1.829e-01  6.725e-02  -2.720  0.01128 *
PERF         4.910e-02  1.781e-02   2.757  0.01032 *
PROTECT      5.238e-01  2.172e-01   2.411  0.02298 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7543 on 27 degrees of freedom
(7 observations deleted due to missingness)
Multiple R-squared:  0.4477,    Adjusted R-squared:  0.3454
F-statistic: 4.377 on 5 and 27 DF,  p-value: 0.004779

```

Figure 1. Output example.

- Compare all observed models and choose the best one.
 - Run your markdown and generate PDF document with all comments and conclusions.
- Upload resulting Rmd and PDF files to the learning environment

OPTIONAL

- Use statistical tests for discovering heteroscedasticity problem (for example, Goldfeld-Quandt test) and autocorrelation problem (for example, Durbin-Watson test).